



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO LAKSHMI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

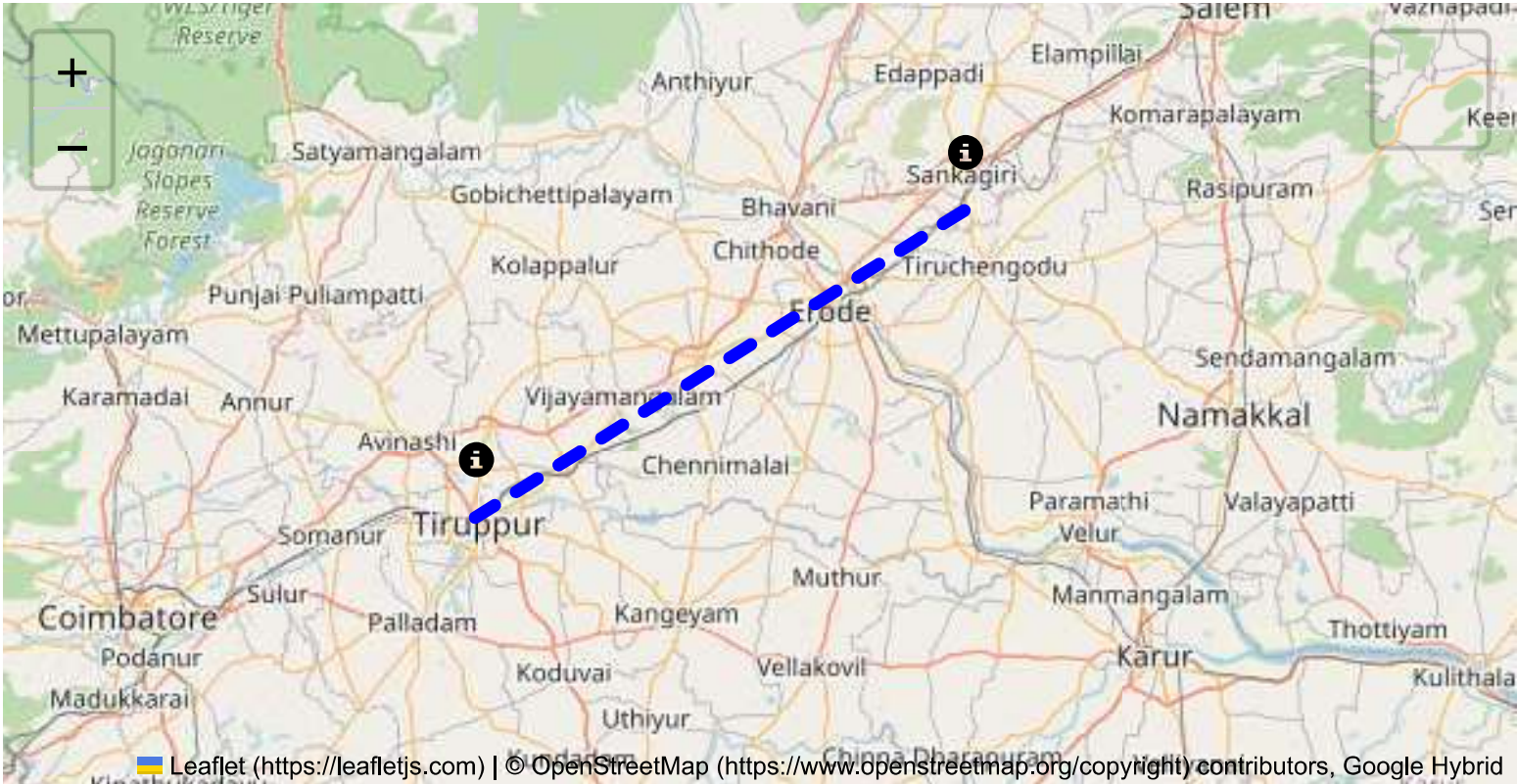
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

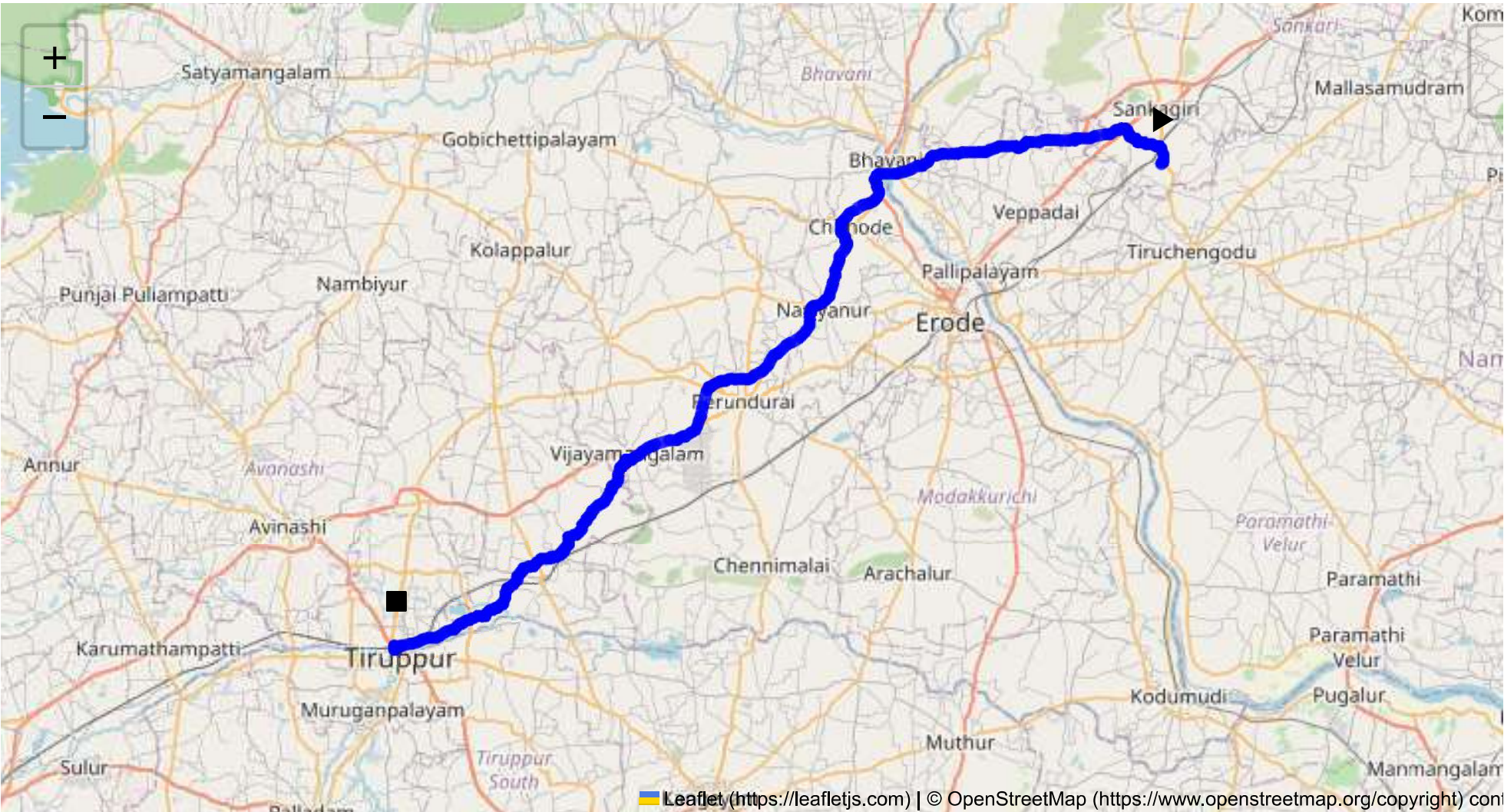
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 82.23 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.109455, 77.340656)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to SF No, 347/3, Tiruppur covers approximately 82.23 kilometers. Primarily, it includes segments of AH47/NH544, a major highway known for its connection between various important industrial hubs in Tamil Nadu. The route is designed to navigate through key towns and

rural areas, with relatively straight paths interspersed with some curvy sections and occasional elevation changes, making it suitable for heavy vehicles.

Typical Weather Conditions and Potential Hazards

The region experiences a tropical climate with hot summers (March to June), a southwest monsoon season (July to September), a northeast monsoon season (October and November), and mild winters (December to February). During monsoon months, heavy rainfall can result in reduced visibility, waterlogged roads, and potential flooding, particularly in low-lying areas. Surface run-off may deposit debris onto the roads, creating slick conditions. Seasonal variations such as strong winds can impact vehicle stability, especially when empty or lightly loaded.

Traffic Patterns and Congestion

- **Peak Hours:** Typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM.
- **Congestion-Prone Areas:** Urban stretches near major towns like Erode and Tiruppur experience higher congestion. Around toll plazas and entry-exit points on NH544, traffic slows down due to vehicle checks and payment queues.
- Access roads to industrial areas may experience heavy congestion, with slow-moving traffic due to the high number of goods vehicles.

Road Quality and Infrastructure

The route primarily follows NH544, which is generally well-maintained with a dual carriageway providing smooth transit. However, some feeder roads could have potholes, uneven surfaces, or inadequate signage. Periodic maintenance work may lead to minor diversions and temporary lane closures.

Alternative Routes for Emergencies

In case of road closures or emergencies:

- Consider using SH 96 or SH 86, which run parallel and can help detour around congestion or accidents.
- Local roads connecting various small towns can serve as bypass options, though these often have less capacity for heavy vehicles.

Regulations Affecting Hazardous Material Transport

Local transportation regulations require:

- Compliance with the Motor Vehicles Act and Hazardous and Other Wastes Rules.
- Drivers to hold a valid license for hazardous material transport, special safety permits, and adherence to designated time slots to minimize public risk.
- Mandatory signage and labeling on vehicles carrying hazardous materials.

Historical Incidents

Past incidents in this region have often involved overturned tanker trucks due to high speed on curves or during adverse weather. There is also a history of minor spills leading to temporary road closures.

Environmental Considerations

- The route crosses numerous agricultural zones which demand attention to how spills can impact soil and water quality.
- Nearby water bodies within 5km of numerous segments necessitate caution to prevent contamination.

Communication Coverage

Most of the route is covered by solid telecommunications infrastructure, providing mobile phone and data coverage. However, certain rural stretches might experience weak signals or brief dead zones, particularly in more isolated areas.

Emergency Response Times

- **Highway Sections:** Approximately 20-30 minutes due to proximity to patrol vehicles and emergency services.
- **Rural Sections:** Response can take between 45 minutes to an hour, subject to weather and road conditions.
- **Urban Areas:** Typically within 15-20 minutes due to local fire and emergency service availability.

Overall Risk Assessment

The route presents moderate risk factors for drivers of heavy vehicles transporting hazardous materials. The primary concerns involve weather conditions, traffic congestion, and environmental sensitivity. Proactive navigation, adherence to safety regulations, and staying informed about road conditions can mitigate these risks effectively. Training in emergency protocols and maintaining up-to-date communication equipment is essential for safe transit along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45509, 77.81383	30 KM/Hr
9	Turn	Medium	11.45085, 77.77020	30 KM/Hr
10	Turn	Medium	11.43165, 77.67734	30 KM/Hr
11	Turn	Medium	11.23724, 77.50686	30 KM/Hr
12	Turn	High	11.23704, 77.50676	15 KM/Hr
13	Blind Spot	Blind Spot	11.23686, 77.50216	10 KM/Hr
14	Turn	Medium	11.23649, 77.50219	30 KM/Hr
15	Turn	High	11.23396, 77.50105	15 KM/Hr
16	Turn	Medium	11.23329, 77.49966	30 KM/Hr
17	Turn	Medium	11.17851, 77.46169	30 KM/Hr
18	Turn	Medium	11.17368, 77.45730	30 KM/Hr
19	Turn	Medium	11.17111, 77.45547	30 KM/Hr
20	Turn	High	11.16959, 77.44396	15 KM/Hr
21	Turn	Medium	11.16381, 77.43700	30 KM/Hr
22	Turn	Medium	11.16280, 77.43139	30 KM/Hr
23	Turn	Medium	11.15023, 77.42247	30 KM/Hr
24	Turn	Medium	11.13848, 77.41611	30 KM/Hr
25	Turn	Medium	11.13578, 77.41299	30 KM/Hr
26	Turn	Medium	11.12989, 77.40302	30 KM/Hr
27	Turn	Medium	11.12465, 77.38649	30 KM/Hr
28	Turn	Medium	11.12232, 77.38291	30 KM/Hr
29	Turn	Medium	11.12104, 77.38220	30 KM/Hr
30	Blind Spot	Blind Spot	11.10798, 77.34071	10 KM/Hr
31	Turn	Medium	11.10746, 77.34098	30 KM/Hr
32	Turn	Medium	11.10736, 77.34122	30 KM/Hr
33	Turn	Medium	11.10748, 77.34161	30 KM/Hr
34	Turn	Medium	11.10764, 77.34174	30 KM/Hr
35	Turn	High	11.10982, 77.34146	15 KM/Hr
36	Turn	Medium	11.10971, 77.34076	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
4	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
5	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
7	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
8	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
9	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
10	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
11	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
12	clinic	Erode Cancer Centre	11.3732, 77.649152	30 km/h	Medium
14	hospital	Irt Hospital	11.2803603, 77.5644118	30 km/h	Medium
16	hospital	Gen Siddha Hospital	11.2375033, 77.5059797	30 km/h	Medium
17	hospital	Vijayamangalm Government Hospital	11.2382827, 77.501687	30 km/h	Medium
19	hospital	Aravind Eye Care Hospital	11.1729469, 77.4574467	30 km/h	Medium
20	clinic	Sri Maha Annai Clinic	11.1707162, 77.4536498	30 km/h	Medium
21	hospital	Ram Doctor Hospital	11.1709391, 77.4532808	30 km/h	Medium
22	clinic	Kumaran Clinic	11.1697553, 77.451543	30 km/h	Medium
23	hospital	Sri Aravindhar Hospital	11.1697756, 77.4448276	30 km/h	Medium
24	clinic	Dr. Raja Clinic	11.1278464, 77.3927768	30 km/h	Medium
25	hospital	Sakthi Sugam Hospital - Tiruppur	11.1151517, 77.3621541	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	hospital	AG Hospital, Tiruppur	11.1090267, 77.3528045	30 km/h	Medium
27	hospital	Nallini Hospital	11.1091685, 77.353517	30 km/h	Medium
28	clinic	S R Nursing Home, Tiruppur	11.111432, 77.3489019	30 km/h	Medium
29	police	TIRUPPUR NORTH	11.107892, 77.3450511	30 km/h	Medium
30	hospital	Tirupur Medical Foundation (P) Limited	11.1090003, 77.3449825	30 km/h	Medium
31	police	Tiruppur North	11.1066254, 77.3427939	30 km/h	Medium
34	hospital	Aadhar Multi-speciality Hospital	11.1113928, 77.3384408	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
6	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
13	college	Government polytechnic college	11.2907053, 77.5698726	30 km/h	Medium
15	school	Bharathi Matriculation School	11.2494613, 77.5307028	30 km/h	Medium
18	college	Sasurie Engineering College	11.2171482, 77.4898552	30 km/h	Medium
32	school	Nanjappa Boys Higher Secondary School	11.1044446, 77.3399245	30 km/h	Medium
33	school	St.Joseph's Matriculation School	11.1046832, 77.3422087	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45509, 77.81383



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45085, 77.77020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43165, 77.67734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23724, 77.50686



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23704, 77.50676



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23686, 77.50216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23649, 77.50219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23396, 77.50105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23329, 77.49966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17851, 77.46169



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17368, 77.45730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17111, 77.45547



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.16959, 77.44396



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16381, 77.43700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16280, 77.43139



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15023, 77.42247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.13848, 77.41611



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.13578, 77.41299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12989, 77.40302



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12465, 77.38649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12232, 77.38291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12104, 77.38220



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.10798, 77.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10746, 77.34098



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10736, 77.34122



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10748, 77.34161



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10764, 77.34174



Coordinates: 11.10982, 77.34146



Coordinates: 11.10971, 77.34076